

Lesson 4.1 – Thinning Cuts

Definition

Removing a branch completely at its point of origin.

Purpose

- Improves air circulation
 - Allows sunlight to penetrate the canopy
 - Reduces overcrowding
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When to Use

- When branches are too dense
 - When removing weak or unwanted growth
 - To open up the center of the tree
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Example

A mango tree with dense inner branches:

- Before thinning → poor airflow and limited sunlight
 - After thinning → better light exposure and healthier growth
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Demo (Step-by-step)

1. Identify overcrowded branches
 2. Select the branch to remove
 3. Locate the branch collar
 4. Cut just outside the collar in one smooth motion
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Lesson 4.2 – Heading Cuts

Definition

Shortening a branch without removing it completely.

Purpose

- Encourages new growth
 - Promotes a bushier plant
 - Controls plant height
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When to Use

- To shape young plants
 - To stimulate lateral growth
 - To manage plant size
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Example

Cutting the tip of a guava branch:

- New shoots grow below the cut
 - Plant becomes fuller and more compact
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Demo (Step-by-step)

1. Select the branch to shorten
 2. Locate a healthy bud
 3. Cut slightly above the bud at a 45° angle
 4. Ensure a clean and smooth cut
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Lesson 4.3 – Comparing Thinning and Heading Cuts

Key Differences

Cut Type | Description | Result

Thinning Cut | Removes entire branch | Opens canopy

Heading Cut | Shortens branch | Stimulates growth

Application Example

In a citrus tree:

- Use thinning cuts → to reduce crowding
 - Use heading cuts → to shape the tree
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Lesson 4.4 – Choosing the Right Technique

Guide

Situation | Recommended Technique
Overcrowded branches | Thinning cut
Excessive height | Heading cut
Weak inner growth | Thinning cut
Need for bushier plant | Heading cut

Scenario

You are pruning a young mango tree:

- Dense interior → apply thinning cuts
 - Long outer shoots → apply heading cuts
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Lesson 4.5 – Common Errors in Pruning Techniques

Mistakes to Avoid

- Using heading cuts too often
→ Causes excessive bushy growth
 - Removing too many branches with thinning cuts
→ Leads to plant stress
 - Incorrect cut placement
→ Slows healing
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Example

Incorrect practice:

- Repeated heading cuts on all branches
→ Results in overcrowded shoots

Correct practice:

- Balanced use of thinning and heading cuts
→ Maintains healthy structure
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Lesson 4.6 – Practice Activity

Activity: Technique Identification

Observe a tree and decide:

- Which branches need thinning cuts
 - Which branches need heading cuts
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Guide Questions

- Is the tree too dense?
 - Are branches too long?
 - Does the plant need shaping or opening?
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Lesson 4.7 – Quick Review

Key Points

- Thinning cuts remove entire branches
- Heading cuts shorten branches
- Use thinning cuts to open the canopy
- Use heading cuts to encourage growth
- Apply techniques properly to avoid plant stress